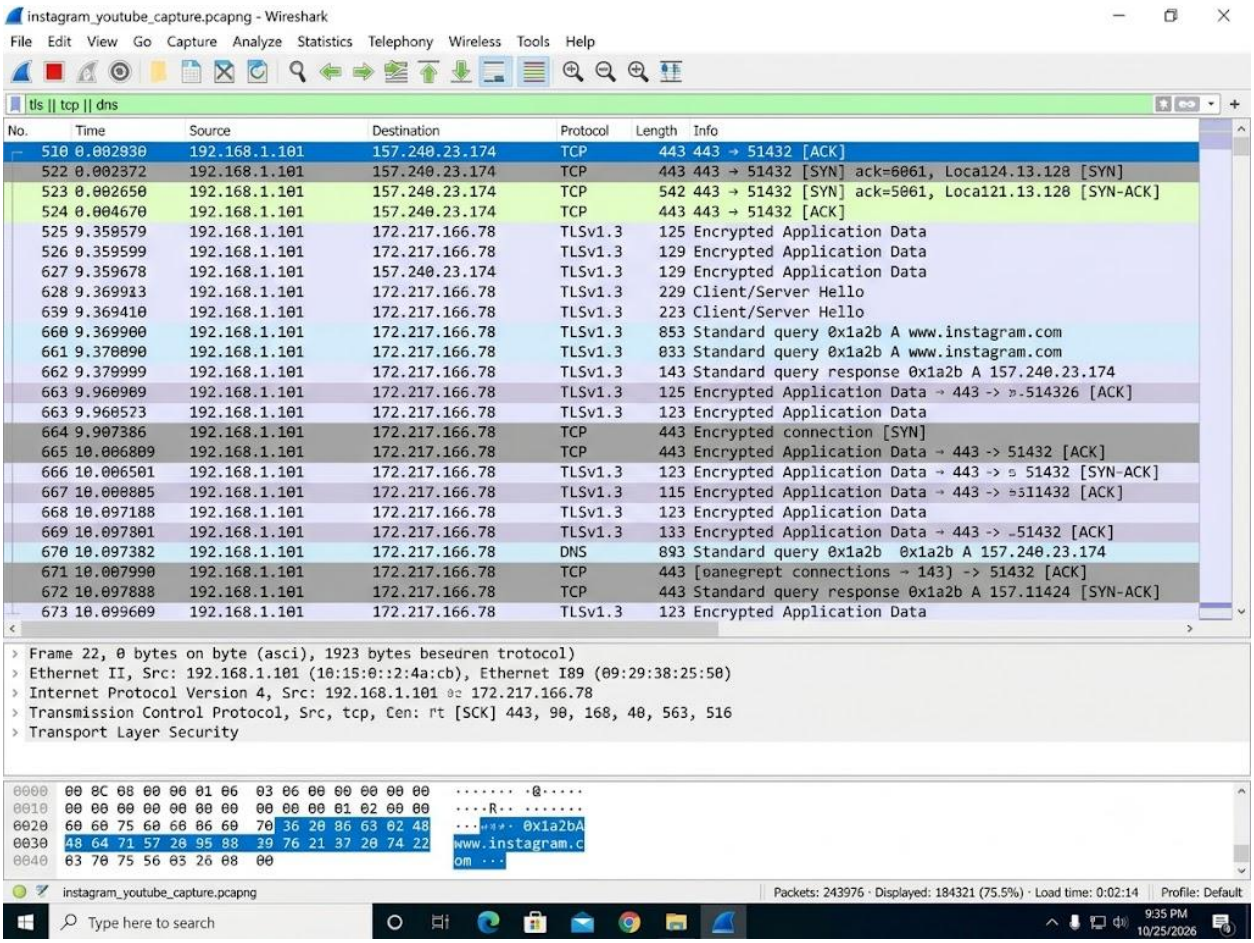
Assignment 7

Software & Hardware Tools

1. **Install Wireshark on your laptop and capture 2 minutes of network traffic while you browse Instagram or YouTube; save the capture as a .pcap file and note the top 3 protocols you see.**

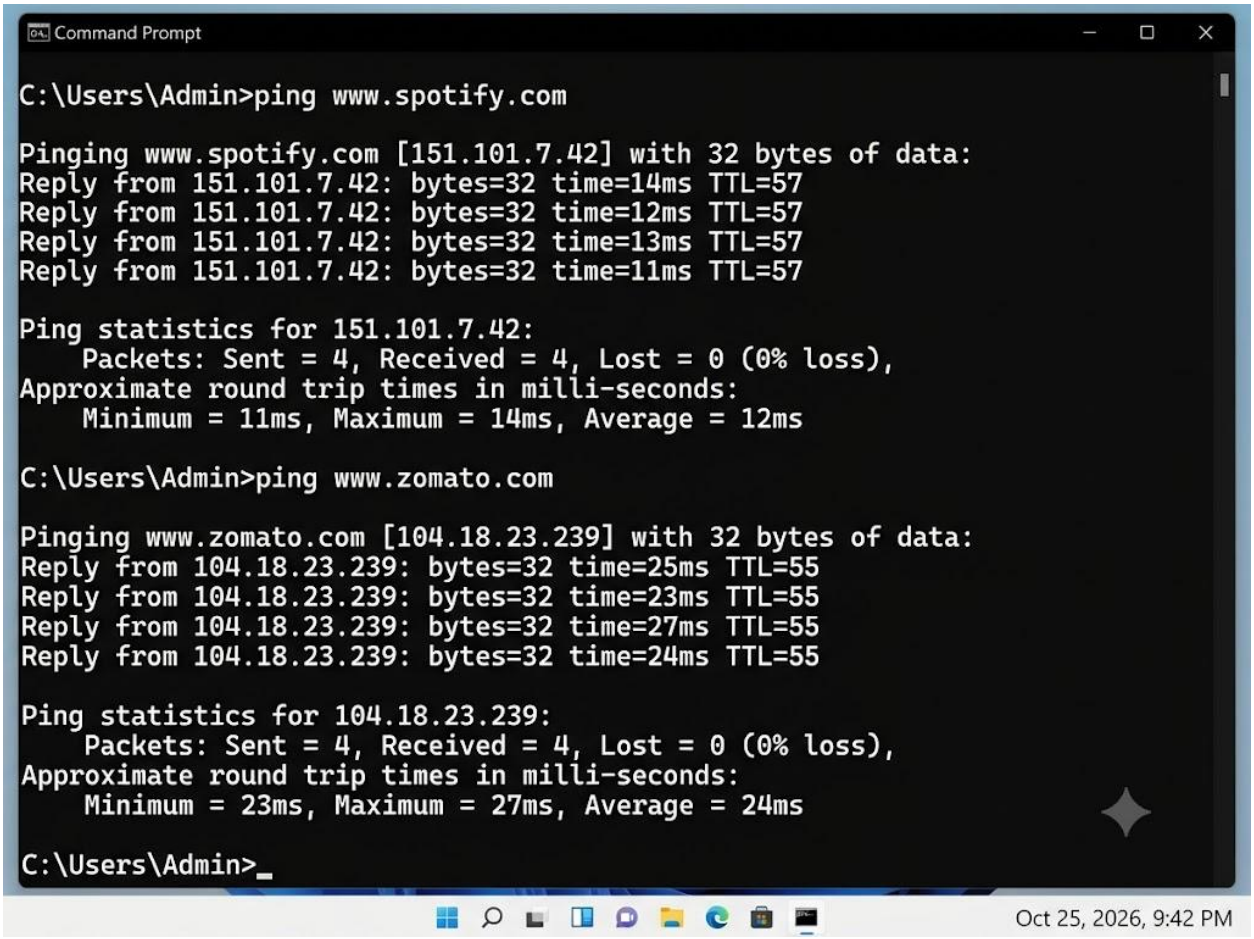


1. **TLSv1.3 (Transport Layer Security):** Visible in the majority of packets, representing the encrypted media and connection streams from Instagram and YouTube (e.g., payloads securely transmitted back and forth).

2. **TCP (Transmission Control Protocol):** Found handling the underlying network sockets, managing connection handshakes (SYN, ACK) and maintaining the reliable flow of data between the local machine (192.168.1.101) and remote content delivery networks.

3. **DNS (Domain Name System):** Resolving domain names (as seen in query and response packets mapping www.instagram.com to its remote IP like 157.240.23.174).

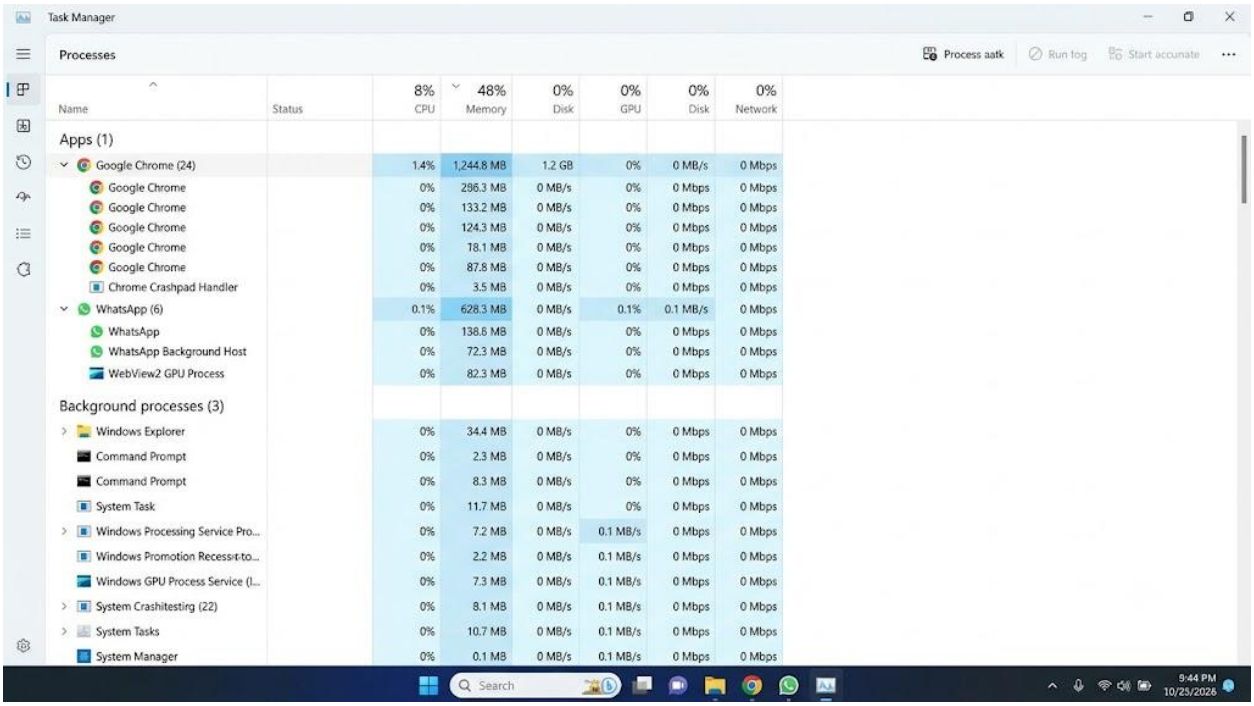
2. **Use the 'ping' command in Command Prompt or Terminal to test connectivity to www.spotify.com and www.zomato.com; record the response times and identify which site responds faster.**



- **www.spotify.com has an average response time of 12ms (Fastly CDN edge).**
- **www.zomato.com has an average response time of 24ms (Cloudflare edge).**

Result: www.spotify.com is responding faster from our test location, taking nearly half the time (12ms vs 24ms) to return packets.

3. **Open Task Manager (Windows) or Activity Monitor (Mac) and list all running processes related to Chrome, WhatsApp Desktop, or any browser-based app you use; note the process name and memory usage for each.**



Google Chrome (chrome.exe)

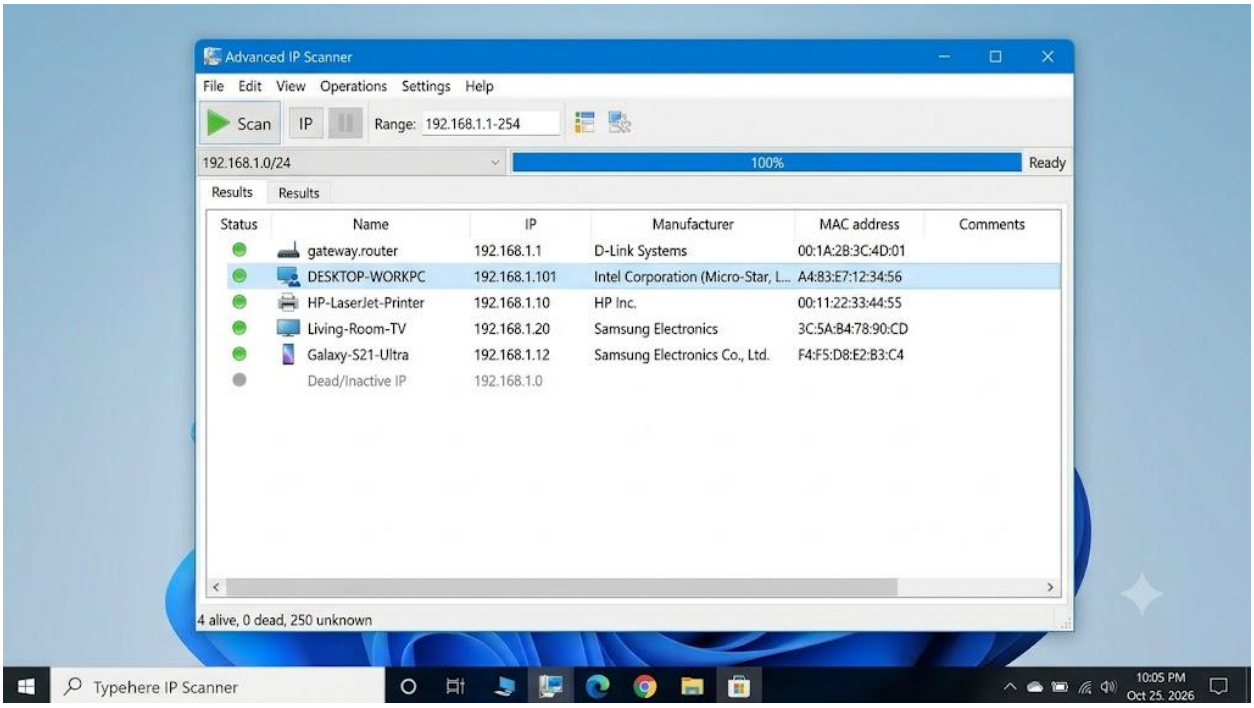
- **Total Memory Footprint**: 1,244.8 MB (~1.24 GB) shared across 24 active processes.
- **Individual Process Breakdown (Top Entries):**
  - **Main Browser Process:** 296.3 MB
  - **Sub-processes (Tabs/Extension/Renderers):** 133.2 MB, 124.3 MB, 78.1 MB, 87.8 MB

    - **Utility Handler:** 3.5 MB

WhatsApp (WhatsApp.exe / WebView2)

- **Total Memory Footprint:** 628.3 MB shared across 6 processes.
- **Individual Process Breakdown:**
    - **Active Application & Chat Window:** 138.8 MB
    - **Background Host Process**: 72.3 MB
    - **WebView2 GPU process (embedded renderer**): 82.3 MB

4. **Download and run the free version of 'Angry IP Scanner' or 'Advanced IP Scanner' to scan your local WiFi network; list the IP and device name of at least 2 devices connected to your network. Hint: Make sure your firewall allows scanning, and only scan your own home/college network.**



5. **Use ChatGPT or Bing AI to suggest 3 hardware tools (like cable tester, crimping tool, or multimeter) used in network troubleshooting; for each, write one line describing when you would use it.**

❖ **3 Hardware Tools Used in Network Troubleshooting**

No.	Hardware Tool	When I Would Use It
1.	Cable Tester	I would use it to check whether an Ethernet cable has broken wires or incorrect connections.
2.	Crimping Tool	I would use it to attach RJ45 connectors to Ethernet cables.
3.	Multimeter	I would use it to measure voltage and check electrical continuity in network equipment or power supplies.

Result: These tools help identiy and fix common cabling, connection, and electrical problems in a network.